

LESSON 8-3 • SESSION 1 • PRACTICE • FORM B

Multiplication Equations

Learning target – I can solve a multiplication equation by dividing both sides by the same number. · 6.EE.B.7

REMEMBER

- The **coefficient** is the number next to the variable. $5b$ means $5 \times b$.
- Division undoes multiplication. **Divide both sides** by the coefficient.
- Check** by multiplying: coefficient $\times$ your answer = the total.

WORKED EXAMPLE

5 boxes each hold the same number of pencils, 45 in all. Solve $5b = 45$.

Step 1 Divide both sides by 5: $5b \div 5 = 45 \div 5$.

Step 2 Simplify: $b = 9$.

Step 3 Check: $5 \times 9 = 45$.

Answer Each box holds 9 pencils.

WORDS **coefficient** (*coeficiente*) · **inverse operation** (*operación inversa*) · **Division Property of Equality** (*propiedad de división de la igualdad*)

PRACTICE

show your work · label units

1. Solve each equation.

a. $13k = 78$ ____

b. $8m = 136$ ____

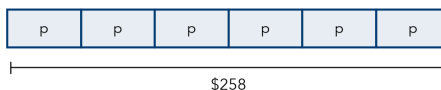
c. $12w = 84$ ____

d. $4y = 14$ ____

e. $16t = 64$ ____

f. $2r = 9$ ____

2. **Soccer Cleats.** A team spent \$258 on 6 pairs of cleats. Each pair cost the same, p dollars.



a. Write a multiplication equation. _____

b. Solve it. _____

c. Check by multiplying. _____

3. **Bookshelf Rows.** A bookshelf holds the same number of books on each of its 8 shelves. It holds 168 books in all. Write and solve an equation for b , the books on one shelf.

Hint Shelves $\times$ books on one shelf = all books.

Show your work.

Answer _____ books

4. **Egg Cartons.** A farm fills 12 cartons with e eggs each, 264 eggs in all. Solve $12e = 264$.

Hint Divide both sides by 12.

Show your work.

e = _____

5. **Training Miles.** A runner ran m miles each day for 16 days, 208 miles in all. Solve $16m = 208$.

Show your work.

m = _____

- 6. Car Wash.** A car wash charges the same price for every car. Washing 28 cars brought in \$420. Write and solve an equation for the price of one wash.

Hint The number of cars is the coefficient.

Show your work.

Price _____

- 7. Field Day Water.** A crew fills 16 identical coolers. Together they hold 224 cups of water. Write and solve an equation for the cups in one cooler. Check it.

Hint Check: $16 \times$ your answer should be 224.

Show your work.

Answer _____ cups

- 8. Concert Rows.** A hall has r rows with 26 seats in each row. It has 754 seats in all.

- Write and solve an equation for the number of rows. _____
- Ten rows are removed for a stage. How many seats are left? _____
- How can you check part a without dividing again? _____

FIND THE ERROR

explain the mistake, then fix it

- 9. Omar's Work.** To solve $6x = 78$, Omar multiplied both sides by 6. He wrote $x = 468$. Explain his mistake. Then solve correctly.

TEST PRACTICE

choose the best answer

- 10.** Solve $7n = 84$.

A $n = 77$ **B** $n = 12$ **C** $n = 588$ **D** $n = 91$

- 11.** To solve a multiplication equation, you –

- multiply both sides by the coefficient
- subtract the coefficient
- divide both sides by the coefficient
- add the coefficient

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

21 I	13 I	29 S	43 D	14 E	\$392 X	22 V	\$15 D	192 Z	1,548 Q
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#2b	#3	#4	#5	#6	#7	#8a
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CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____